Dataset Replay Report

Purpose: isolate **policy vs. execution** for the "cube not grasped, arm too high" failure. We replay the *exact recorded joint targets* from a training episode (no policy in the loop) and compare what the robot measures against what the dataset recorded.

- If replay-measured motion **matches** the dataset recording (and the cube is grasped) → execution/sim2real is faithful → the failure is in the **policy**.
- If the replay itself **diverges** (e.g. ends too high) → the gap is in **execution** (sim2real dynamics) → then re-run replay with TAM on to see if it closes that specific deviation.

Setup

Dataset	`~/dataset/compliant_test_flip_pnp_softer/blocks_success`
Control	joint PD, **soft** `kp=40, kd=13` (pd_mode=0, matches the dataset)
Command	`action.joint_target` (absolute rad), replayed at 25 fps
Clamp	`max_relative_movement = 20°` (targets jump up to ~21°/step; see notes)
Gripper	inverted (dataset 0/1 → live 0=closed/1=open)
TAM	**off** for the first pass (bare execution gap), then on
Script	`examples/inference/franka_replay.py`
Plot	`examples/inference/plot_replay.py`

Data notes (from loading episode 0)

- 370 frames (14.8 s). `pd_mode` is all-0 (soft) as expected.
- **The terminal frame has `action.is_valid=False` and a garbage target** (130° jump on J4/J6) — dropped by the loader.
- Recorded **targets jump up to ~21°/step** while the recorded **observed motion is ≤1.8°/step** — the compliant controller ran heavily error-saturated (target far ahead, arm lagging). The 20° clamp therefore barely engages (~1 frame).

How to reproduce

```
# TAM off (bare execution gap):
python examples/inference/franka_replay.py
# then plot dataset vs measured:
python examples/inference/plot_replay.py tam_logs/<ts>

# TAM on (does it close the gap?): set ReplayConfig.tam = True, re-run, re-plot.
```

Outputs per run in `tam_logs/<ts>/` : `replay_frames.csv` (per-frame `q_sent` / `q_meas` / `q_ds` + gripper), `replay_{tam,notam}_controller_1khz.csv` (1 kHz controller/TAM detail), `replay_meta.json` (episode/config provenance), `replay_compare.png` (the plot).

Results — episode 0, TAM off (`20260821_161209`)

The dataset is SIMULATION (TAMP planner: `outputs/compliant_test/flip_then_pnp` , domain/problem PDDL, per-episode domain-randomization metadata). So `q_ds` is the sim-achieved motion and `q_meas` is the real robot running the same targets — **this replay measures the sim2real execution gap directly**, with TAM off.

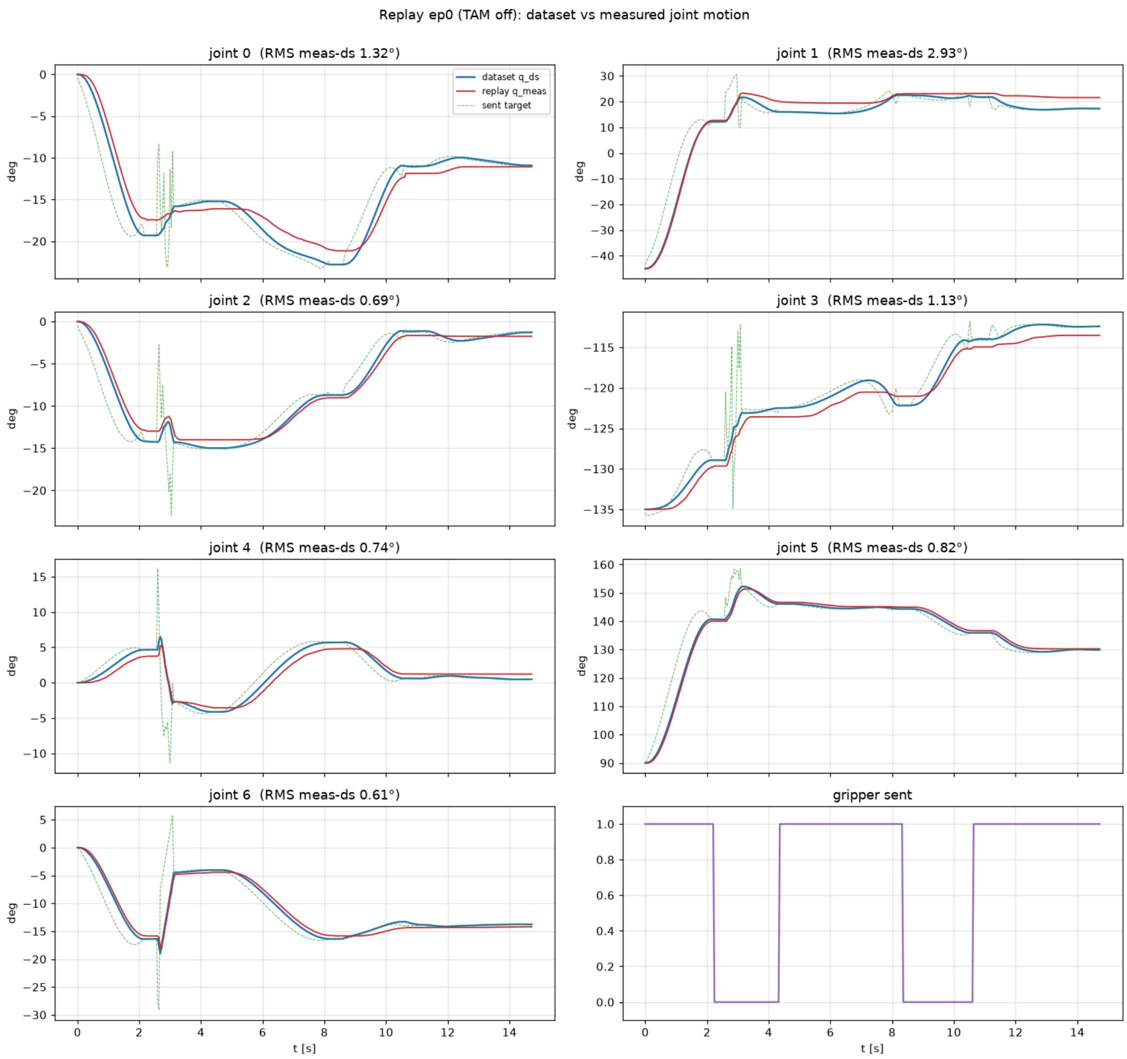
369 frames, 14.7 s.

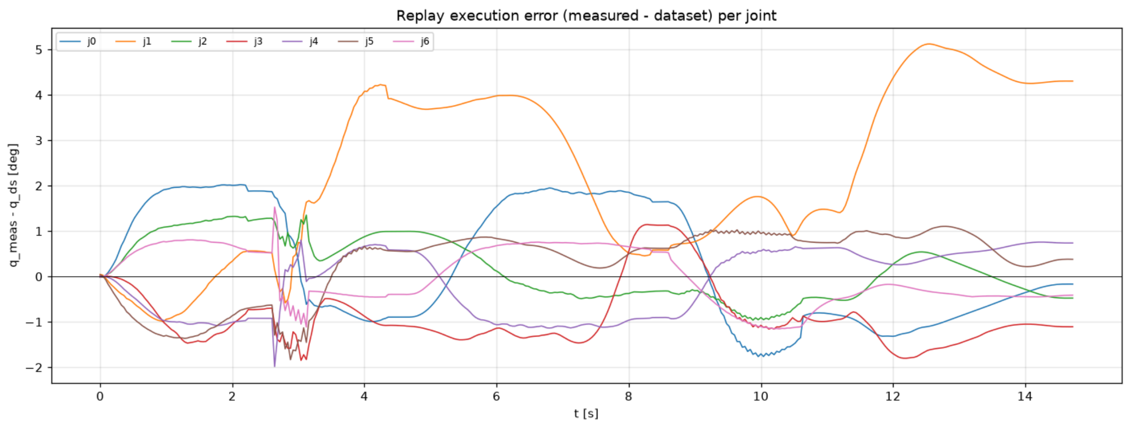
Joint space — replay tracks the sim joint trajectory to a few degrees

RMS(`q_meas` − `q_ds`) per joint [deg]:

j0	j1	j2	j3	j4	j5	j6
1.32	**2.93**	0.69	1.13	0.74	0.82	0.61

max | `q_meas` − `q_ds` | per joint [deg]: `[2.0, 5.1, 1.4, 1.9, 2.0, 1.8, 1.5]` (worst is J1, the shoulder).



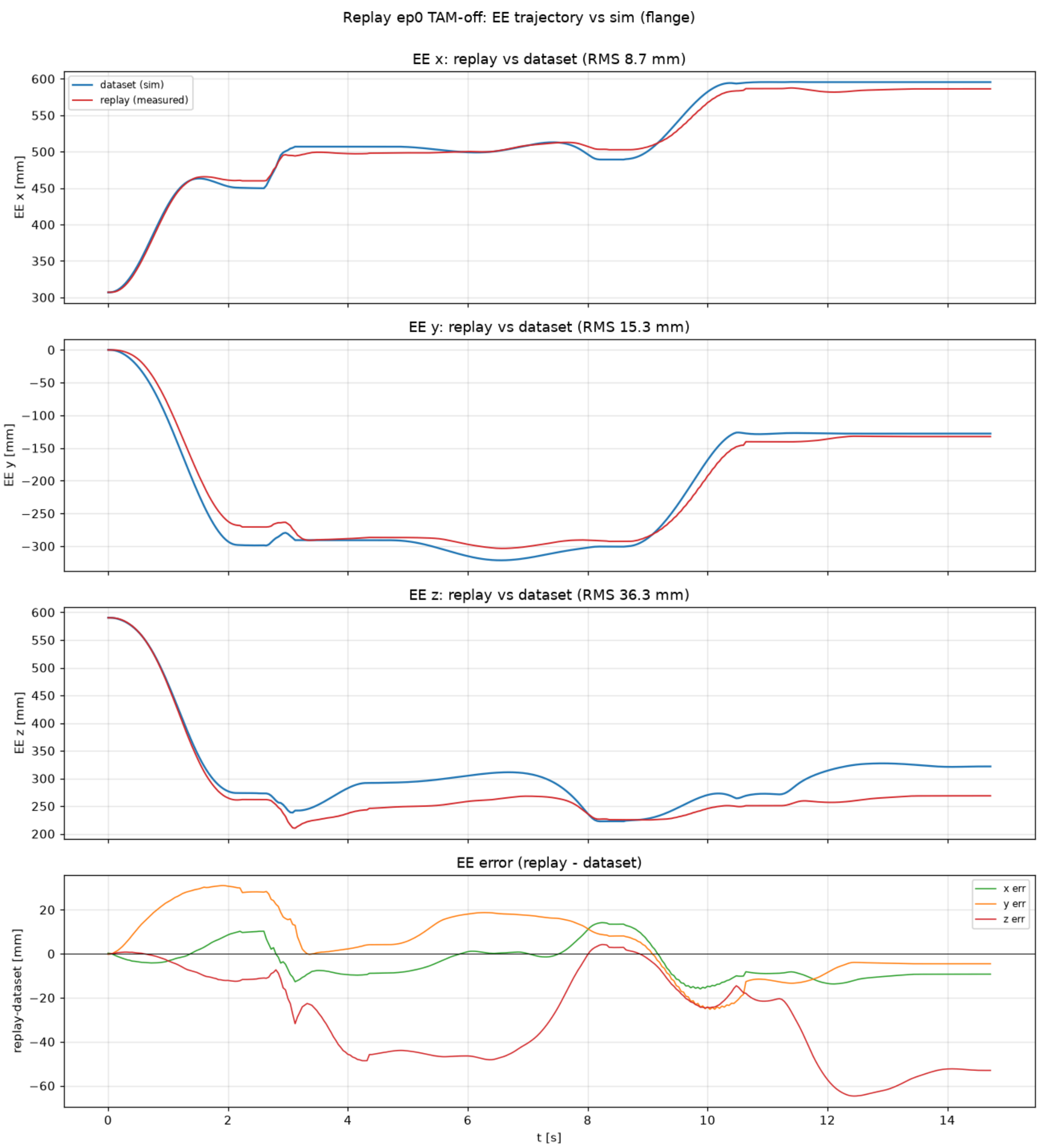


EE space — small joint errors amplify to grasp-relevant cm-level EE error

Forward kinematics (FR3, **flange** frame — matches the dataset EE convention and the robot's `tcp_offset=identity`) of `q_meas` vs `q_ds`:

axis	RMS [mm]	max |·| [mm]
x	8.7	16.0
y	15.3	31.0
z (height)	**36.3**	**64.7**

Mean EE-Z(`replay` − `sim`) = **−29.7 mm** (replay runs ~3 cm *lower* on average), but the vertical error swings up to **6.5 cm** peak.



Verdict

- **Execution is NOT grasp-faithful.** Even replaying the planner's exact targets, the real robot's EE deviates from the sim by up to **6.5 cm (36 mm RMS), dominated by the vertical axis** — comparable to a cube's size, so enough to miss a grasp. The gap is small in joint space (1–3°) but amplifies through the kinematics.
- This is precisely the **sim2real execution gap TAM is meant to close**, and it is present with **TAM off**. So the "arm too high / cube not grasped" symptom is at least partly an **execution** problem, not purely the policy. (Note the sign: this replay averages ~3 cm *lower* than sim; the live policy looked *higher* — the two aren't directly comparable since the policy commands its own targets, but both point to an uncorrected vertical execution error.)

Results — episode 0, TAM on (`20260821_162326`) vs off

Same episode, TAM enabled (stable config: `tam_residual_clip=[…,0.5,0.5,0.5]` , `rate_limit=True` , `tam_history_pre_ratelimit=True`). TAM ran **stably** through the whole replay: `active=0.97` , OOD max **2.3σ**, wrist J7 `dq` ≤0.73 rad/s, latent age median 99 ms — no limit cycle.

TAM improves joint tracking and EE x/y...

Joint RMS(`q_meas` – `q_ds`) [deg]:

	j0	j1	j2	j3	j4	j5	j6
TAM off	1.32	2.93	0.69	1.13	0.74	0.82	0.61
TAM on	**0.59**	**2.38**	**0.43**	1.28	**0.35**	**0.44**	**0.41**

EE deviation from sim [mm] (flange frame):

axis	RMS off	RMS on	max off	max on
x	8.7	**7.6**	16.0	**14.2**
y	15.3	**9.6**	31.0	**21.8**
z	36.3	**33.9**	64.7	**51.9**
3D	40.4	**36.0**		

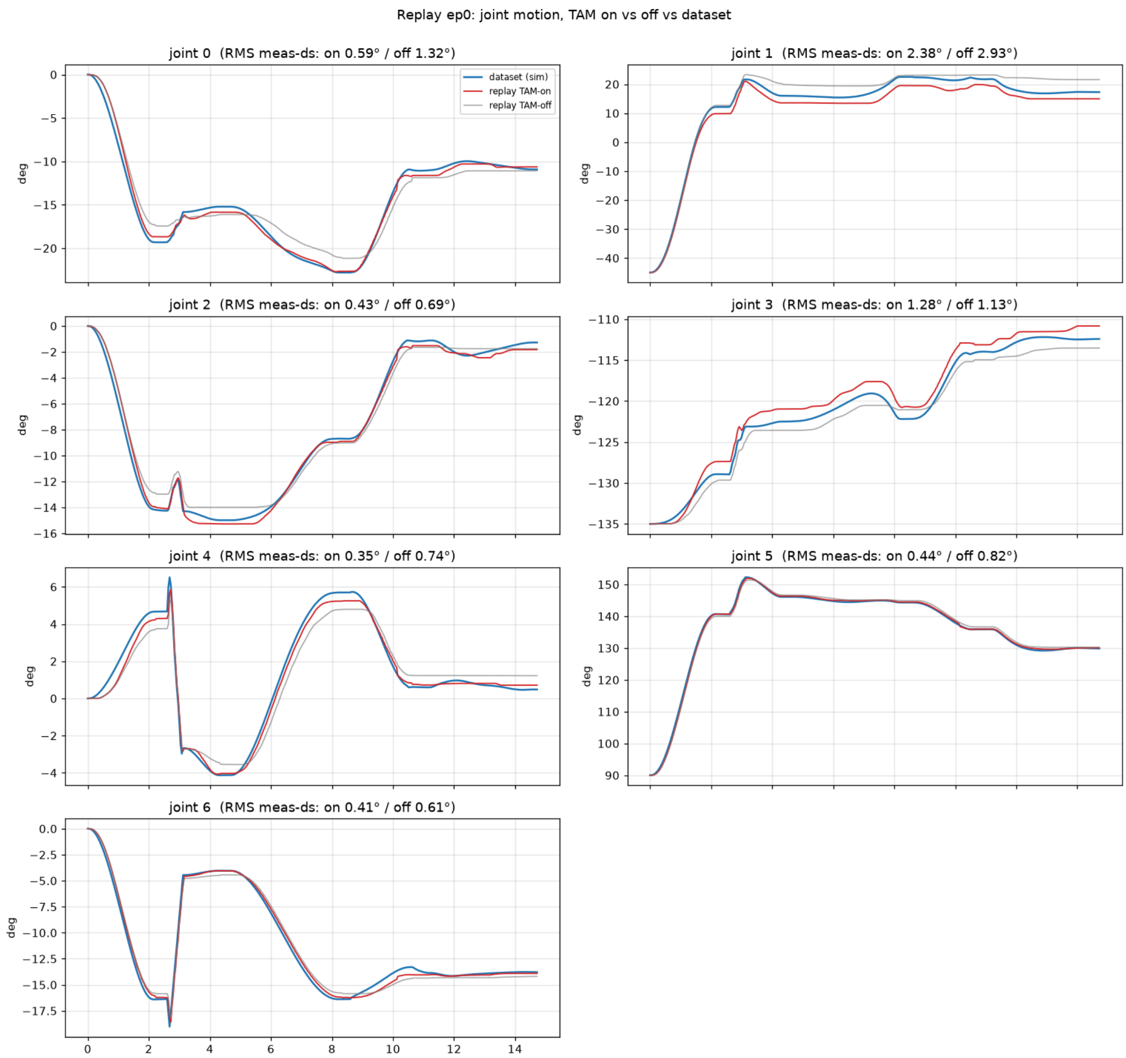
TAM improves the **y** error (~37%), modestly improves x and the 3D total, and improves joint tracking on 6 of 7 joints — evidence it is closing part of the sim2real dynamics gap, and doing so stably.

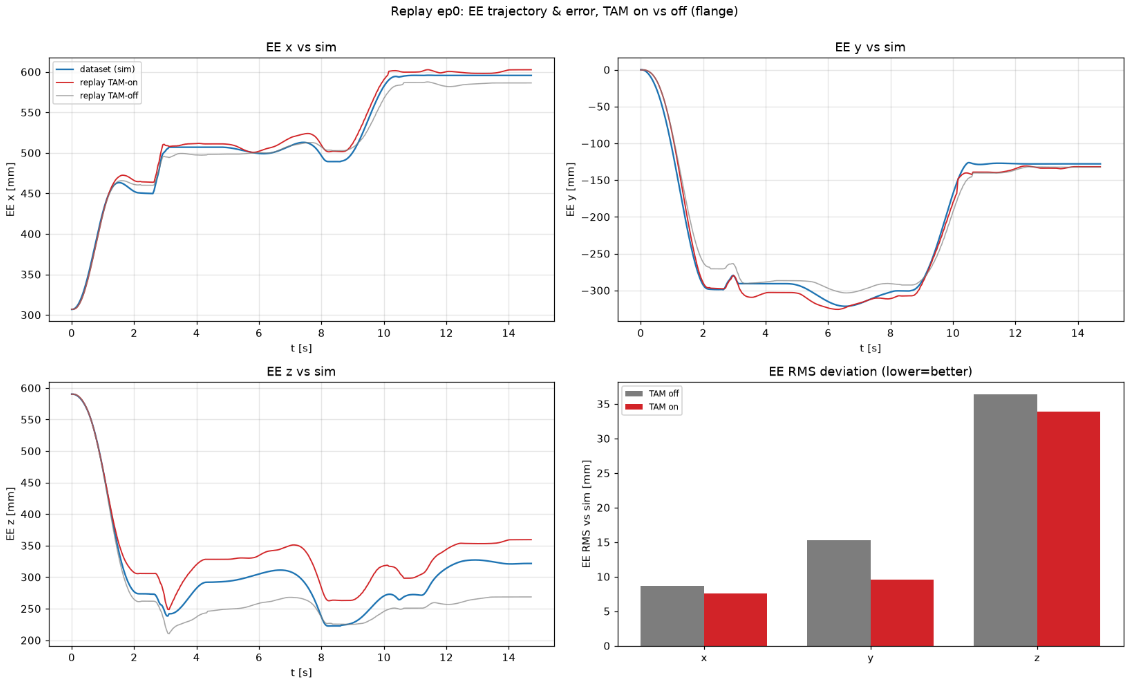
...but it does NOT fix the vertical (grasp-critical) axis — it overcorrects it

The Z RMS barely moves (36→33 mm), and the **mean EE-Z bias flips sign**:

	mean EE-Z (replay – sim)
TAM off	**–29.7 mm** (arm ~3 cm too *low*)
TAM on	**+31.5 mm** (arm ~3 cm too *high*)

So TAM adds ~+60 mm of vertical correction and **overshoots** — turning a 3 cm *undershoot* into a 3 cm *overshoot*. **This matches the observed live symptom ("arm too high, cube not grasped"):** with TAM on, the EE sits ~3 cm above where the sim demo put it, which is enough to miss the cube.


Replay ep0: joint motion, TAM on vs off vs dataset
joint 0 (RMS meas-ds: on 0.59° / off 1.32°)
dataset (sim)
replay TAM-on
replay TAM-off
joint 1 (RMS meas-ds: on 2.38° / off 2.93°)
joint 2 (RMS meas-ds: on 0.43° / off 0.69°)
joint 3 (RMS meas-ds: on 1.28° / off 1.13°)
joint 4 (RMS meas-ds: on 0.35° / off 0.74°)
joint 5 (RMS meas-ds: on 0.44° / off 0.82°)
joint 6 (RMS meas-ds: on 0.41° / off 0.61°)
deg



Verdict

- **TAM works and is stable** — it closes the dynamics gap in joint space and in EE x/y, without reigniting the wrist instability.
- **The grasp failure is a persistent ~3 cm vertical bias, not a TAM stability problem.** Both directions show a ~30 mm systematic EE-Z offset; TAM flips its sign (over-corrects) rather than removing it. A constant few-cm vertical bias that survives in both cases smells like a **calibration/model offset**, not a dynamics gap TAM can learn away.

Next steps

1. **Chase the ~3 cm vertical bias directly** (independent of TAM): check the **TCP offset / gripper mounting**, the **payload/load mass** used for gravity compensation, the **FK model vs. real** (this Z error appears in the bare soft controller too), and **table-height** calibration. A constant offset here would explain both signs.
2. **Understand TAM's vertical overshoot** — it adds ~2× the needed Z correction. Consider whether the shoulder/elbow (J1/J3, which drive EE-Z) residuals are too aggressive; a per-joint scale on those could center the Z error near zero.
3. Confirm grasp timing given the **inverted gripper** before attributing remaining failures.

> *Caveat: gripper was replayed **inverted** (dataset→live convention); confirm open/close timing before trusting grasp outcome. EE numbers are FK (FR3 + Robotiq 2F-85 TCP) of measured vs. sim joints — a wrong TCP offset would bias the EE-Z comparison itself (see step 1).*

Dataset 2 — `aug13_sim2real_kp40_pnp` (pnp-only, kp40, sim)

Same two experiments on a second **simulation** dataset (pnp-only task, soft kp40). Episode 0, 262 frames. Baseline `20260821_163425` (TAM off, rerun after the first attempt drove into the table), TAM-on `20260821_163735`.

TAM-on ran **stably**: `active=0.95`, OOD max **1.8σ**, wrist J7 `dq` ≤0.58 rad/s, latent age median 96 ms — no limit cycle (wrist clip config unchanged).

The vertical-overshoot finding reproduces

Joint RMS(`q_meas` – `q_ds`) [deg] — TAM improves 6 of 7 joints:

	j0	j1	j2	j3	j4	j5	j6
TAM off	1.37	2.65	0.69	1.08	0.82	0.75	0.86
TAM on	**0.55**	**2.53**	**0.51**	1.24	**0.42**	**0.34**	**0.62**

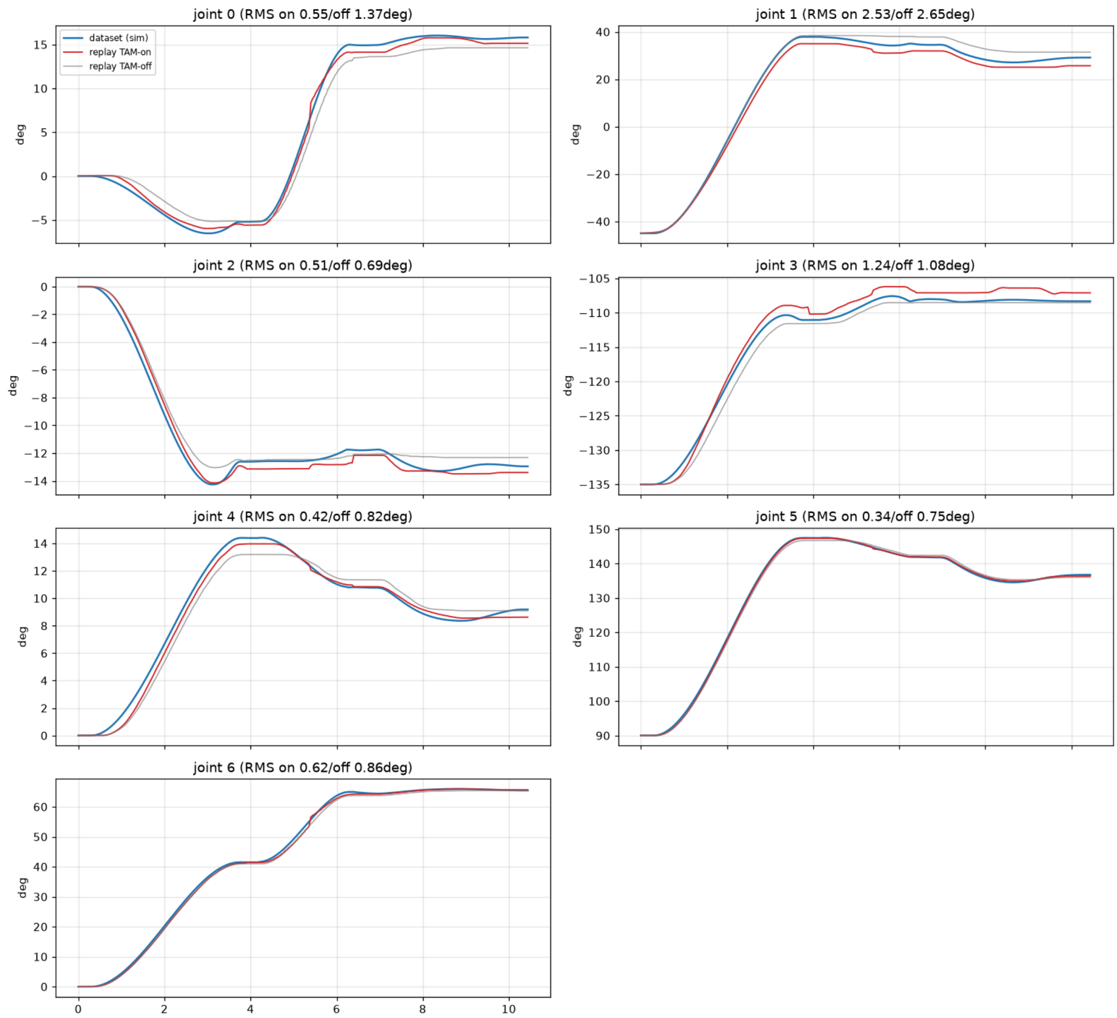
EE deviation from sim [mm] (flange):

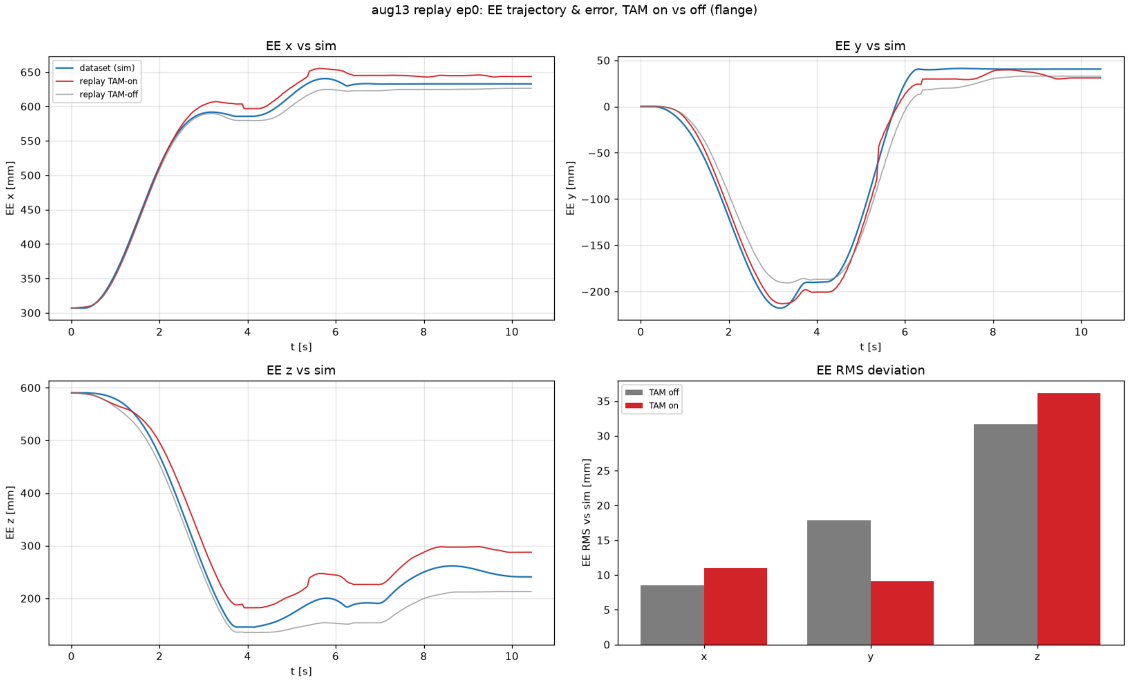
axis	RMS off	RMS on	max off	max on
x	8.5	11.0	17.0	19.2
y	17.8	**9.1**	29.7	**16.4**
z	31.7	36.1	52.0	50.6
3D	37.3	38.8		

Mean EE-Z bias flips sign again — same as dataset 1:

	mean EE-Z (replay – sim)	EE-Z bottoms out at
TAM off	**–27.7 mm** (too *low*)	135 mm — **10 mm below sim (145 mm)** → **into the table**
TAM on	**+31.9 mm** (too *high*)	182 mm — **37 mm above sim** → **cannot reach the cube**

aug13 replay ep0: joint motion, TAM on vs off vs sim





Cross-dataset conclusion

The vertical execution bias and TAM's response are **consistent across both datasets**:

	compliant_test	aug13_kp40_pnp
mean EE-Z, TAM off	−29.7 mm (low)	−27.7 mm (low)
mean EE-Z, TAM on	+31.5 mm (high)	+31.9 mm (high)

- **TAM off → arm ~3 cm too low** (drives into the table on the pnp dataset).
- **TAM on → arm ~3 cm too high** (bottoms out 37 mm above sim → misses the cube). This is exactly the live "arm too high, cube not grasped" symptom.
- TAM reliably improves joint tracking and the **y** axis and stays stable, but it **over-corrects the vertical axis by ~2×**, converting a −3 cm undershoot into a +3 cm overshoot rather than centering it. On aug13 the net EE 3D RMS is even slightly worse (37.3→38.8 mm) because the z overshoot dominates and x worsens.

The blocker is a systematic ~3 cm vertical offset, seen with TAM off, that TAM flips instead of removes. This is very likely a **calibration/model** issue (TCP/flange, payload/gravity comp, FK-vs-real, or table-height), not something TAM can learn away — pursue the vertical-bias hunt (report step 1) before expecting TAM to fix the grasp. A per-joint down-scale on the shoulder/elbow (J1/J3) residuals that drive EE-Z could also stop TAM from overshooting.

Dataset 2 with added payload — block mass in gripper

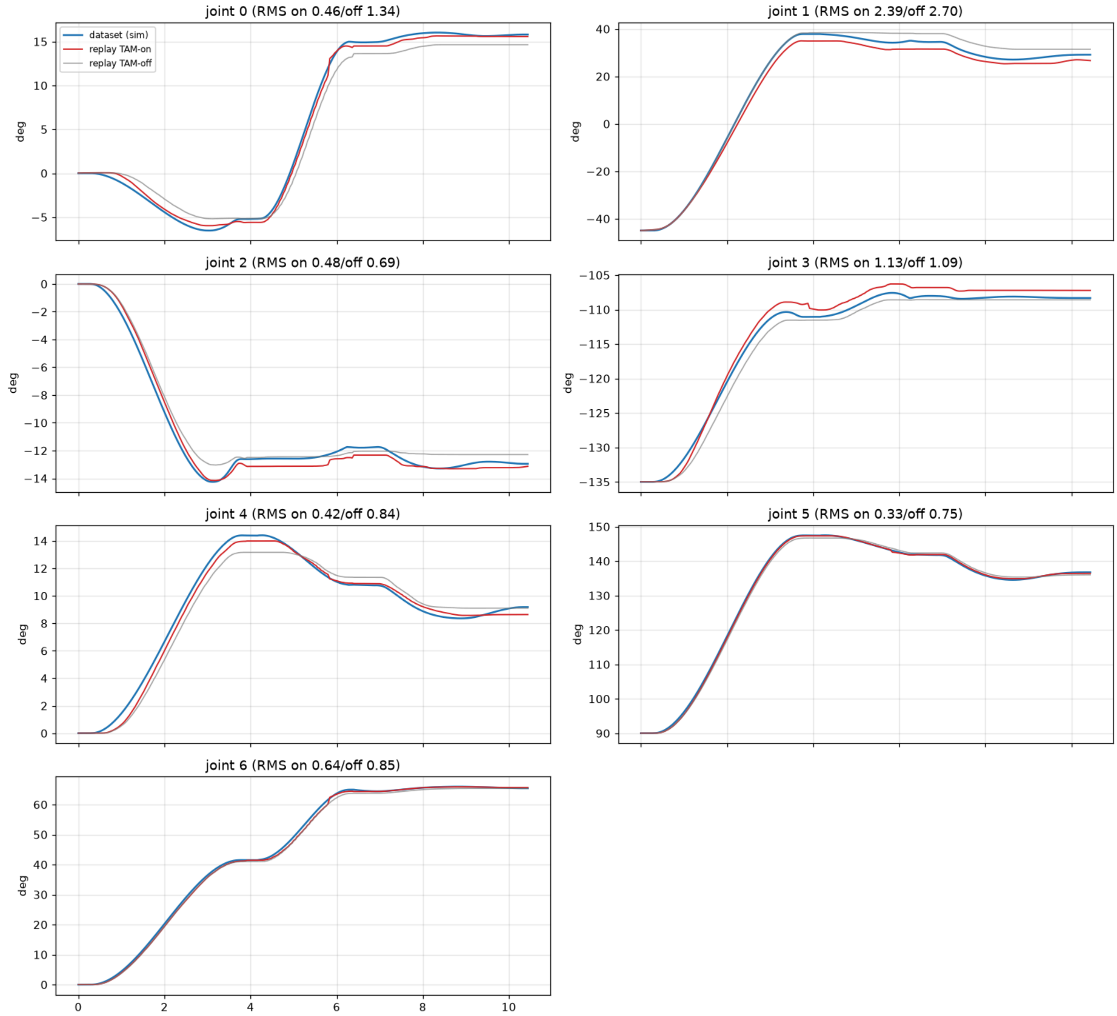
Same aug13 ep0 replay, but with a physical **block placed in the gripper** to add mass. TAM off `20260821_164347`, TAM on `20260821_164438`. (Note: the replay does not call `setLoad`, so the controller's gravity compensation does **not** know about the block — an unmodeled payload.)

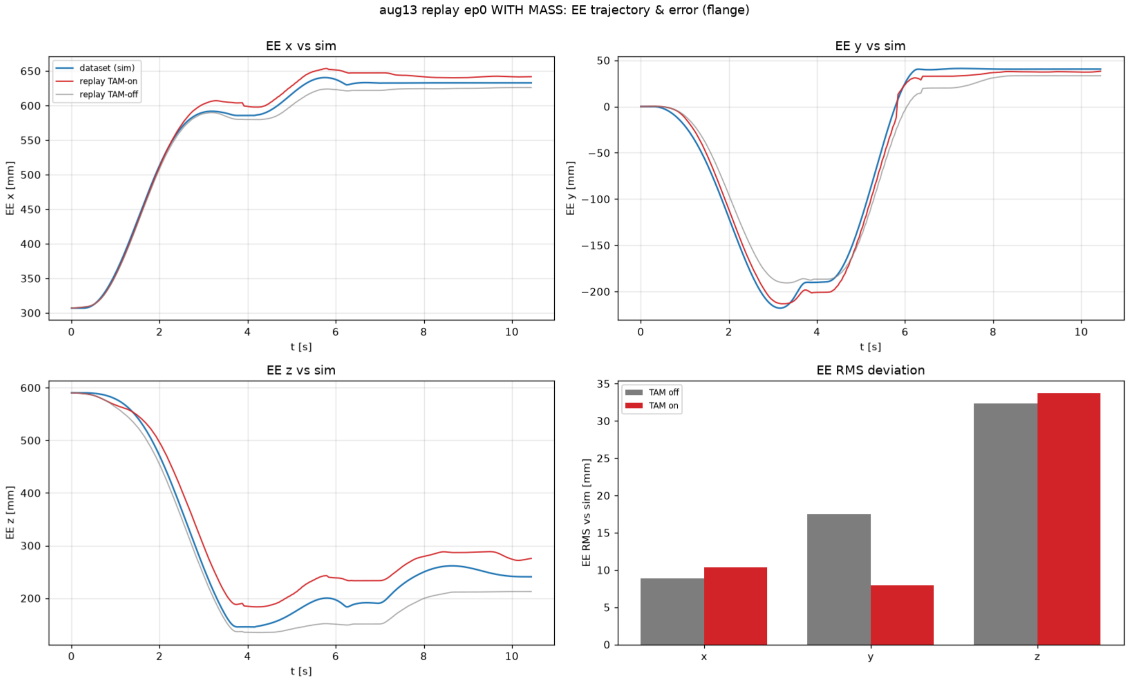
EE-Z bias, with vs without the added mass:

run	mean EE-Z (replay – sim)	EE-Z bottoms at	joint RMS J1 [deg]
no-mass, TAM off	–27.7 mm	135 mm	2.65
mass, TAM off	**–28.3 mm**	135 mm	2.70
no-mass, TAM on	+31.9 mm	182 mm	2.53
mass, TAM on	**+29.7 mm**	184 mm	2.39

TAM-on with mass stayed **stable** (OOD max 1.8σ, J7 `dq` ≤0.58, active 0.95).

aug13 replay ep0 WITH MASS: joint, TAM on vs off vs sim





Finding — the vertical bias is payload-independent

Adding the block **did not change** the results: EE-Z bias, the bottom-out height, and per-joint tracking are all within noise of the no-payload runs (off −28.3 vs −27.7 mm; on +29.7 vs +31.9 mm).

This is an informative **negative result**. If the ~3 cm vertical offset were driven by an **unmodeled payload / gravity-compensation** error, adding a block (which the controller's gravity comp ignores) should have made the soft arm sag noticeably *lower* with TAM off. It did not. So:

- **The ~3 cm vertical bias is NOT a gravity/load-modeling problem** — it is robust to payload.
- That points the calibration hunt (report step 1) toward a **payload-independent kinematic/model offset**: TCP/flange definition, FK-model-vs-real, mounting, or table-height reference — rather than load mass or gravity compensation.
- TAM's vertical **overshoot to ~+3 cm reproduces with payload too**, unchanged.